Heart Disease Prediction Using Machine Learning Algorithms

A comparative study of classification models

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Problem Statement, Motivation & Objectives

Problem statement

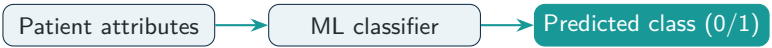
Predict whether a patient belongs to the positive or negative heart-disease class using demographic and clinical attributes.

Motivation

- Heart disease risk is associated with multiple patient measurements.
- Machine learning can learn patterns from these attributes.
- Comparing classifiers helps assess the effect of model choice.

Project objectives

- Explore the dataset and target distribution.
- Use 13 attributes to predict the target class.
- Train eight classification algorithms.
- Compare models using cross-validation and multiple test metrics.



Dataset Description

Dataset at a glance

Total records	303
Input features	13
Target classes	2
Missing values	6 across 2 features
Training records	242 (80%)
Test records	61 (20%)

Target variable

- 0: absence of heart disease
- 1: presence of heart disease

(original target values 1–4)

Predictor features

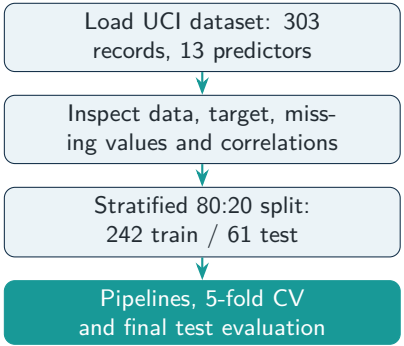
Feature	Meaning
age	Age of the patient
sex	Sex (0/1 coded)
cp	Chest-pain type
trestbps	Resting blood pressure
chol	Serum cholesterol
fbs	Fasting blood sugar indicator
restecg	Resting ECG result
thalach	Maximum heart rate achieved
exang	Exercise-induced angina
oldpeak	Exercise-related ST depression
slope	Slope of peak exercise ST segment
ca	Major vessels observed by fluoroscopy
thal	Thalassemia / thal test category

Data Understanding & Experimental Workflow

Initial data analysis

- Dataset shape: 303 × 14 (13 predictors + target).
- Positive class (1): 139 records (45.87%).
- Negative class (0): 164 records (54.13%).
- Stronger absolute correlations: thal, ca, exang, oldpeak, thalach and cp.
- Weakest absolute correlation: fbs.

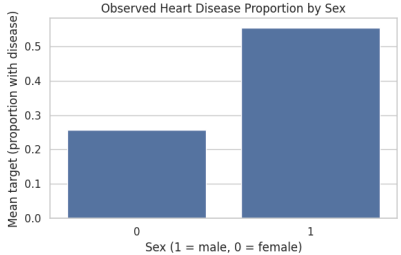
Project workflow

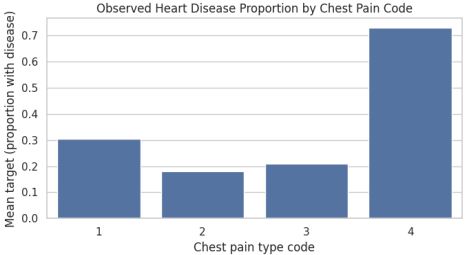


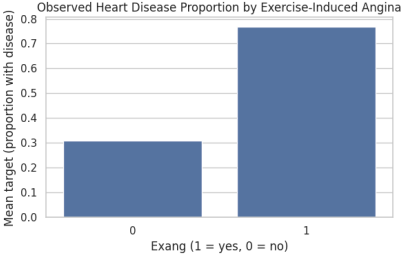
Implementation note

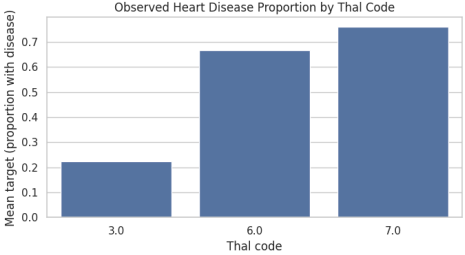
UCI data has 4 missing values in `ca` and 2 in `thal`; pipelines apply median imputation and scaling where needed. Stratified split (seed 42); five-fold CV uses training data only.

Exploratory Data Analysis: Key Observations









Bar height = observed proportion with disease in each category (not correlation).

Machine Learning Models Compared

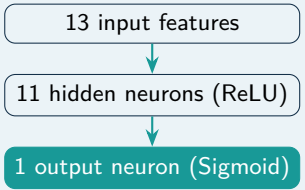
Classical classifiers

- **Logistic Regression**
  Linear probabilistic classifier.
- **Gaussian Naive Bayes**
  Conditional-independence model.
- **Linear SVM**
  Maximum-margin linear boundary.
- **K-Nearest Neighbors**
  Neighbor voting with k = 7.

Tree-based models

- **Decision Tree**
  Feature-based decision rules.
- **Random Forest**
  Ensemble of decision trees.
- **XGBoost**
  Gradient-boosted trees for binary classification.

Neural network architecture



MLPClassifier with Adam optimizer, maximum 1,500 iterations and early stopping.

Why compare different families?

The models represent linear, probabilistic, distance-based, tree-ensemble, boosting and neural approaches.

Mathematical Foundation: Binary Classification

From patient attributes to a prediction

For patient i, the 13 clinical attributes form a feature vector:

$$\mathbf{x}_i = [x_{i1}, x_{i2}, \ldots, x_{i,13}]^T, \quad y_i \in \{0, 1\}.$$

A linear model computes a score:

$$z_i = \mathbf{w}^T \mathbf{x}_i + b.$$

Logistic regression maps this score to a probability:

$$\hat{p}_i = \sigma(z_i) = \frac{1}{1 + e^{-z_i}}.$$

The predicted class is 1 when $\hat{p}_i \geq 0.5$, and 0 otherwise.

Binary cross-entropy loss

The neural network is trained to minimise:

$$L_{BCE} = -\frac{1}{n} \sum_{i=1}^{n} [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)].$$

- y_i: actual class (0 or 1).
- $\hat{p}_i$: predicted probability of class 1.
- $n = 242$: training observations in this split.

Intuition: confident correct predictions incur low loss, while confident incorrect predictions incur a large penalty.

How Do Some of the Models Make Decisions?

Neural network used in this project

Architecture: $13 \to 11 \to 1$.

$h = \mathrm{ReLU}(W_1 x + b_1), \quad \hat{p} = \sigma(W_2 h + b_2).$

ReLU is $\max(0, u)$; sigmoid maps output to $[0, 1]$. Training uses Adam optimization and early stopping (maximum 1,500 iterations).

K-nearest neighbors (KNN)

Euclidean distance:

$$d(x, x^{(j)}) = \sqrt{\sum_{k=1}^{13} (x_k - x_k^{(j)})^2}.$$

Predict the majority label among the $k = 7$

Decision tree: Gini impurity

For class proportions p_0 and p_1 in a node:

$$Gini = 1 - (p_0^2 + p_1^2).$$

Choose a split that reduces the weighted Gini impurity of its child nodes.

Random forest: combining trees

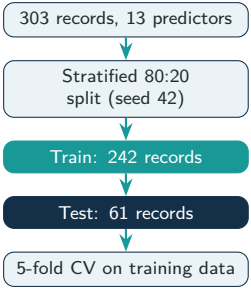
Each tree predicts $h_t(x)$. The forest aggregates the predictions:

$$\hat{y} = \mathrm{mode}\{h_1(x), \ldots, h_T(x)\}.$$

The final class is the majority vote across trees.

Experimental Setup & Evaluation Method

Train-test protocol



Evaluation protocol

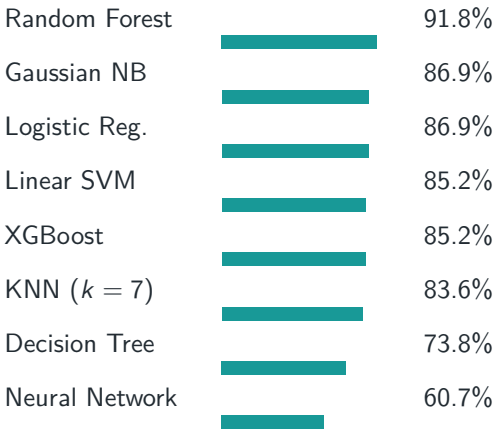
Eight models are compared using five-fold stratified cross-validation on training data. Final predictions on the held-out test set are evaluated using accuracy, precision, sensitivity/recall, specificity, F1-score and ROC-AUC.

Model selection and final evaluation

Model comparison uses five-fold stratified cross-validation on the training set. The held-out test set is reserved for the final evaluation of each fitted model.

Results: Test Accuracy Across Eight Models

Accuracy on the 61-observation test set



Best observed result

91.8%

Random Forest

56 correct predictions out of 61.

Key comparisons

- Gaussian Naive Bayes and Logistic Regression: 86.9% each.
- Linear SVM and XGBoost: 85.2% each.
- Neural Network: 60.7% on this split.

Single stratified split; five-fold cross-validation is used for development comparison.

Discussion: Interpreting the Results

Observed performance

Random Forest: 91.8% (56 of 61 test cases) - highest test accuracy.

Gaussian Naive Bayes and Logistic Regression: 86.9% each.

Linear SVM and XGBoost: 85.2% each.

KNN: 83.6%; Decision Tree: 73.8%; Neural Network: 60.7%.

Cross-validation comparison

Mean training CV ROC-AUC: Gaussian Naive Bayes 0.897; Logistic Regression 0.896; Linear SVM 0.892; Random Forest 0.882.

Limitations

- Only 303 records from one dataset.
- A single held-out test split; results may vary.
- No independent external validation.
- Medical data may contain population and measurement biases.
- Calibration and clinical utility are not assessed.

Why this matters

Random Forest leads test accuracy; Gaussian Naive Bayes leads mean CV ROC-AUC. In screening, review false negatives, sensitivity, specificity and calibration - not accuracy alone.

Conclusion & Future Improvements

Conclusion

Developed a binary classification workflow using 13 patient attributes and 303 records.

Compared eight models spanning linear, probabilistic, distance-based, tree, boosting and neural approaches.

Random Forest achieved 91.8% test accuracy; Gaussian Naive Bayes ranked highest by mean cross-validation ROC-AUC.

Model choice and evaluation method both affect the conclusions.

Future improvements

- Validate on larger, independent datasets.
- Tune hyperparameters only inside training folds.
- Report confidence intervals and calibration.
- Assess subgroup performance and clinical utility.
- Work with domain experts on thresholds and error costs.

Take-home message

Independent validation is still needed before making generalisation claims.

Educational use only - not a clinical diagnostic tool.

Thank You!

Questions & Discussion

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